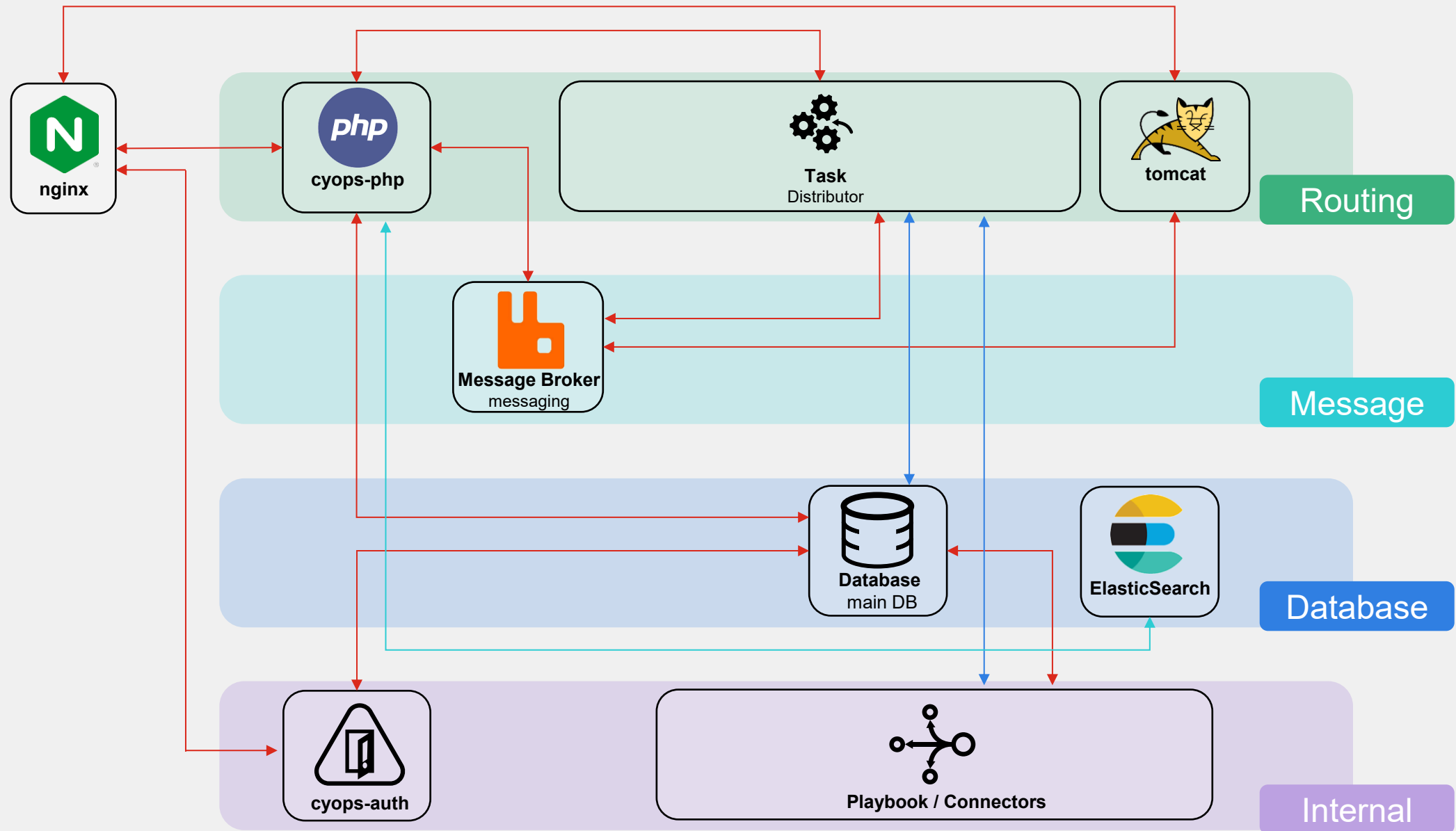




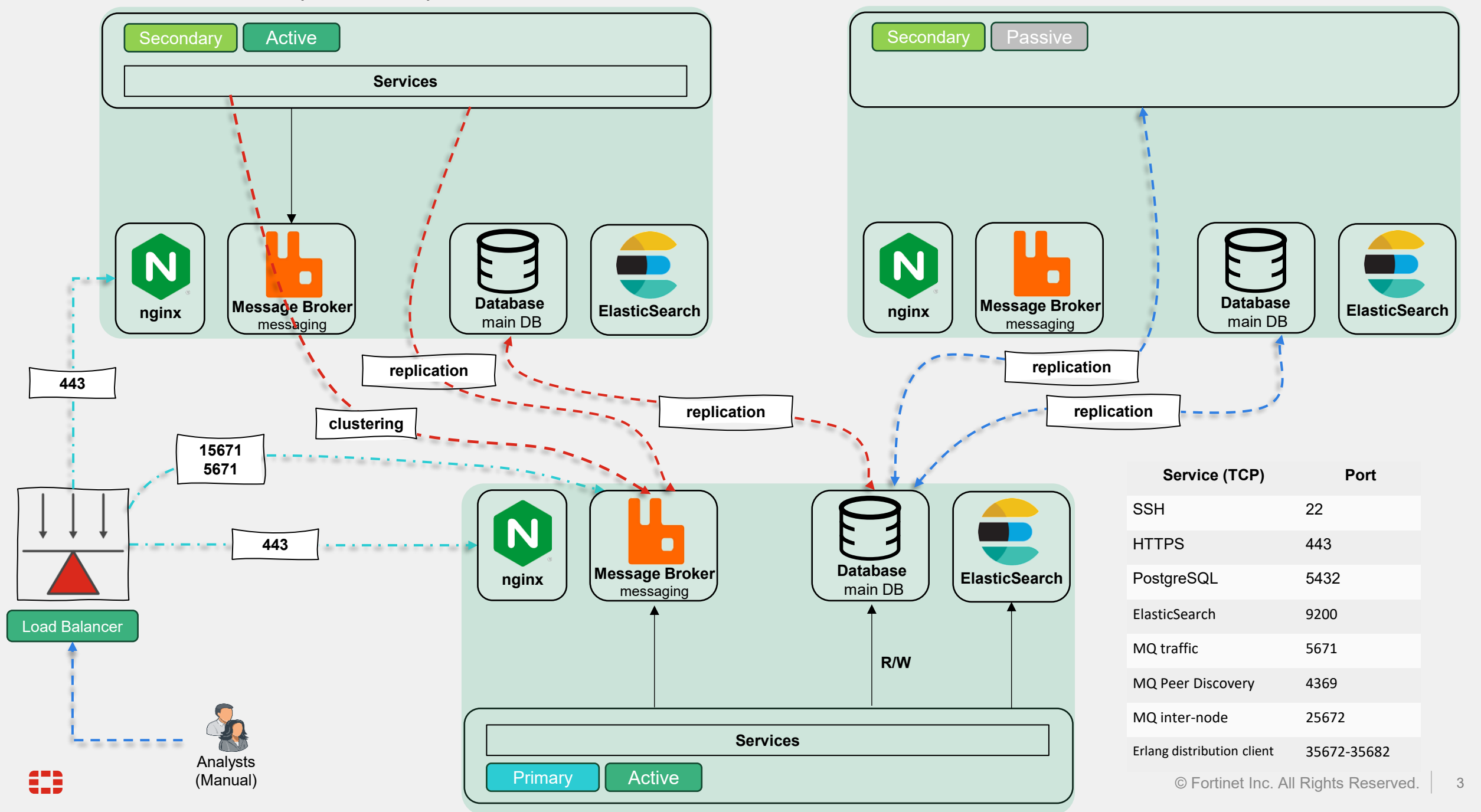
FortiSOAR

Architecture and Trouble Shooting

FortiSOAR Internal Services (Minified)



FortiSOAR Internal Services (HA minified)



FortiSOAR Services

nginx	Main web server handling URL routing
php-fpm	App server handling REST API and all CRUD requests it's the maestro for all the other components
postgresql	Service to Database mapping: Modules & Records → venom Playbooks → sealab Connectors → connectors Tomcat (Audit) → gateway Tomcat (Live sync) → notifier Postman → postman cyops-auth → das
elasticsearch	The engine behind global search functionality, data is always synced between Postgres and Elastic
uwsgi	Gateway service between php-fpm and Workflows/Integrations and Rule Engine (notification engine). Needs restart each connector code change
celeryd/celerybeatd	celeryd fetches the jobs (playbooks) to run from RabbitMQ celery queue and executes them asynchronously. Celerybeatd at the other hand runs the scheduled playbooks
cyops-auth	Service used to generate access token, handles LDAP authentication when configured
cyops-tomcat	Service used for SSO (SAML), auditing by writing events in db, and websocket getting user activity and sending notifications
cyops-search	Service used for full-text searching, finding similar records and predicting the value of fields based on similarity (when ML engine is not in use)
rabbitmq-server	Service is responsible to transfer messages/jobs between the different components of FortiSOAR it's involved in sending audit and live sync notifications, playbook jobs, integration tasks...and many more inter process communication. This service is also responsible for data transfer in a distributed environment (Agent/Dedicated tenant communication)
cyops-postman	Service is responsible for setting up and managing FortiSOAR's distributed multi-tenant setup.
cyops-integrations-agent	Service is responsible for running actions on remote FortiSOAR agents.
cyops-ha	Service is responsible for setting up High Availability in the FortiSOAR environment

Where to Find Logs

Log Name	Purpose
/var/log/messages	General system and services logs
cyops-api/ssl_cyops_api_access.log	Used for troubleshooting web (nginx) UI or API access issues.
/var/log/panopta-agent/panopta-agent.log	FortiMonitor agent logs
cyops-api/ssl_cyops_api_error.log	Used for troubleshooting API errors.
cyops-api/prod.log	Used for troubleshooting FortiSOAR PHP related issues.
cyops-api/last_system_publish.log	Used for troubleshooting publishing issues. It captures the output when a publish is fired from the UI after MMD changes.
cyops-auth/das.log	Used for troubleshooting FortiSOAR user authentication issues.
cyops-auth/fdn.log	Used for troubleshooting FortiSOAR license synchronization issues with FortiGuard Distribution Network (FDN).
cyops-auth/ha.log	Used for troubleshooting FortiSOAR High Availability issues.
cyops-gateway/auditlog.log	Used for troubleshooting FortiSOAR audit log issues.
cyops-gateway/etl.log	Used for troubleshooting FortiSOAR Tomcat application and system-level issues.
cyops-gateway/saml.log	Used for troubleshooting FortiSOAR SAML issues.
cyops-search/falcon.log	Used for troubleshooting FortiSOAR Global Search issues.
cyops-gateway/gateway.log	Used for troubleshooting FortiSOAR Gateway issues such as, audit log page failing to load or the SSO (SAML) configuration page failing to load.



Where to Find Logs

Log Name	Purpose
cyops-notifier/notifier.log	Used for troubleshooting FortiSOAR Web Socket issues (Tomcat).
cyops-workflow/beat.log	Used for troubleshooting issues of the FortiSOAR Scheduler handled by celerybeatd
cyops-workflow/celeryd.log	Used for troubleshooting FortiSOAR playbook runtime issues.
cyops-workflow/sealab.log	Used for troubleshooting FortiSOAR playbook framework issues.
cyops-workflow/ssl_cyops_workflow_access.log	Used for troubleshooting playbook access issues.
cyops-workflow/ssl_cyops_workflow_error.log	Used for troubleshooting playbook errors.
cyops-workflow/uwsgi.log	Used for troubleshooting FortiSOAR playbook and connector issues.
cyops-integrations/connectors.log	Used for troubleshooting FortiSOAR connector related issues.
cyops-integrations/integrations.log	Used for troubleshooting FortiSOAR connector framework issues.
cyops-integrations/integrations/imap/listener.log	Used for troubleshooting the IMAP connector issues.
cyops-integrations/ssl_cyops_integrations_access.log	Used for troubleshooting connector access issues.
cyops-integrations/ssl_cyops_integrations_error.log	Used for troubleshooting connector errors.
csadm/db.log	Used for troubleshooting database externalization errors.
\$version.log for example, 7.6.0-964.log.	Used for troubleshooting FortiSOAR installation issues. Install logs are named according to the FortiSOAR version and build number.
install/connectors.log	Used for troubleshooting connector installation issues.
upgrade_fortisoar_<version_number>-<timestamp>.log	Stores upgrade console log and you can use it to troubleshoot FortiSOAR upgrade issues.
install/config-vm-<timestamp>.log	Used for troubleshooting FortiSOAR Configuration Wizard issues.



Control Log Level

For **sealab** or **workflow**:

1. Open the `/opt/cyops-workflow/sealab/sealab/config.ini` file and set the `WORKFLOW_LOG_LEVEL` parameter to the required logging level. For example, `WORKFLOW_LOG_LEVEL = 'INFO'`
2. Restart the `uwsgi` service.

For **integrations (Connectors)**:

1. Open the `/opt/cyops-integrations/integrations/configs/config.ini` file and set the `connector_logger_level` parameter to the required logging level. For example, `connector_logger_level= 'INFO'`
2. Restart the `uwsgi` service.

For **celeryd**:

1. Open the `/etc/celery/celeryd.conf` file and set the `CELERYD_LOG_LEVEL` parameter to the required logging level. For example, `CELERYD_LOG_LEVEL = 'INFO'`
2. Restart the `celeryd` service.

For **nginx (UI), API, or php**:

1. Open the `/opt/cyops-api/app/config/config_prod.yml` file and set the `level` parameter to the required logging level. For example, `level = 'INFO'`
2. Run the `# systemctl restart php-fpm nginx` command.



Restart Services

- To restart all services at once:

- `csadm services -restart`

- To restart Individual service:

- `systemctl restart SERVICE_NAME`

- If an error occurs use:

- `systemctl status SERVICE_NAME` and `journalctl -xf` to get the error messages



Troubleshoot Connectors

- While: `tail -f /var/log/cyops/cyops-integrations/connectors.log`
- Configure the connector if your issue not getting an “available” status
- Run the action which is not working either with a playbook or with the Action menu @UI
- If the log level is not enough, you may increase it as seen before:
 - `connector_logger_level= 'INFO' at /opt/cyops-integrations/integrations/configs/config.ini`
- If that is still not enough you could also add `logger.warning("message" or variables)` to your `operations.py` of the connector you are troubleshooting to see which values the connector is returning. You will need to restart uwsgi and clear the connector pycache after each modification:
 - `cd /opt/cyops-integrations/integrations/connectors/CONNECTOR_VERSION`
 - `rm -rf __pycache__/ && systemctl restart uwsgi`



Database Troubleshooting

- Typical DB issues should be reported when the clients (almost every service) try to do some operations on the database
 - To connect to any database: `psql -U cyberpgsql -d DB_NAME`
 - To list tables: `\dt`
 - To list fields of a table: `\d+ TABLE_BNAME`
 - Changing Postgres worker memory:
 - When the primary data in the system becomes large (eg, over million alerts) and you notice that the system is slow to respond. The slowness could be caused due to database queries taking longer with the increased database size.
 - You can fine tune this behavior by increasing the following Postgres settings based on the available free memory on the system:
 - Increase the shared buffer `shared_buffers = 2048MB` and worker memory `work_mem = 16MB`
- in `/var/lib/pgsql/16/data/postgresql.conf` then restart Postgres `systemctl restart postgresql-12`



Client-Side Troubleshooting

- Open DevTool (Edge/Chrome)
- Browse to Network

The screenshot displays the Chrome DevTools Network tab. The top toolbar includes options like 'Preserve log', 'Disable cache', and 'No throttling'. The filter bar is set to 'Fetch/XHR'. The 'Show overview' checkbox is checked. The network timeline shows a single request. The selected request is expanded, showing the following details:

- Name:** alerts?\$limit=2147483647&\$orderby=...
- General:**
 - Request URL:** https://34.89.79.14:14005/api/query/alerts?\$limit=2147483647&\$orderby=%2Btype.orderIndex,%2Bstatus.orderIndex
 - Request Method:** POST
 - Status Code:** 200 OK
 - Remote Address:** 34.89.79.14:14005
 - Referrer Policy:** strict-origin-when-cross-origin
- Response Headers:**
 - Cache-Control: no-cache, private
 - Connection: keep-alive
 - Content-Encoding: gzip
 - Content-Type: application/json
 - Date: Thu, 26 May 2022 04:37:52 GMT
 - Link: <https://34.89.79.14:14005/api/3/docs.jsonld>; rel="http://www.w3.org/ns/hydra/core#apiDocumentation"
 - Server: nginx
 - Strict-Transport-Security: max-age=63072000; includeSubdomains; preload
 - Transfer-Encoding: chunked
 - X-Content-Type-Options: nosniff
 - X-Frame-Options: SAMEORIGIN
 - X-XSS-Protection: 1; mode=block
- Request Headers:**
 - Accept: application/json, text/plain, */*

The bottom of the screenshot shows the Console tab with an error message:

```
Uncaught TypeError: Cannot read properties of undefined (reading 'runtime')
at Module.<anonymous> (AdsContent.js:1:6170)
at c (AdsContent.js:1:124)
at AdsContent.js:1:923
at AdsContent.js:1:932
```

Client-Side Troubleshooting

- Open DevTool (Edge/Chrome)
- Browse to Network
- Open Timing TAB

The screenshot displays the Chrome DevTools interface with the Network tab selected. A list of network requests is shown on the left, with 'files?path=installed' highlighted. The Timing tab is open on the right, showing a detailed breakdown of the request's duration. A red box highlights the 'Request/Response' section, which includes 'Request sent' (297.33 ms), 'Waiting (TTFB)' (177.76 ms), and 'Content Download' (0.35 ms). The total duration for the request is 760.18 ms. Below the timing breakdown, the 'Explanation' section provides additional context. At the bottom of the screen, the Console tab shows an 'Uncaught TypeError: Cannot read properties of undefined (reading 'runtime')' error, with the stack trace pointing to 'AdsContent.js:1:6170'.

Category	Sub-category	Duration
Request/Response	Request sent	297.33 ms
	Waiting (TTFB)	177.76 ms
	Content Download	0.35 ms
Total		760.18 ms

```
Uncaught TypeError: Cannot read properties of undefined (reading 'runtime')
    at Module.<anonymous> (AdsContent.js:1:6170)
    at c (AdsContent.js:1:124)
    at AdsContent.js:1:923
    at AdsContent.js:1:932
```



Client-Side Troubleshooting

- Open DevTool (Edge/Chrome)
- Browse to Application
- Under Local Storage select FortiSOAR
- Add the key/value: DEBUG:true
- This will enable all JS console errors in the browser

The screenshot shows the Chrome DevTools Application tab. The left sidebar is expanded to 'Local Storage' for the URL 'https://34.89.79.1...'. A red box labeled '2' highlights the 'Local Storage' section. The main pane shows a list of storage items. A red box labeled '1' highlights the 'Application' tab, and a red box labeled '3' highlights the 'DEBUG' key in the list. The value for 'DEBUG' is 'true'. Below the storage list, the 'Frames' section shows 'top'. At the bottom, the 'Console' tab is active, showing several log messages. The first message is 'Authentication failed!' with a red error icon. The console also shows messages from 'angular.js' and 'chrome-extension://bbcinkgjjkejfdpemiealijmmooekmp/onloadwff.js'.

Key	Value
DEBUG	true
persist.cs.previousState	
cs.TOKEN	"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1dWlkjoiYVQyNTRjMDgtNzFhZS00MTMzLWE1NTA1NTZlNDI0MzQy..."
cs.user_type	2
cs.authentication_configTOKEN	
cs.lastActive	1653505963
cs.system_settings	
cs.modules	
cs.metadata.actors	
cs.metadata.attribute_metadatas	
cs.metadata.avatars	
cs.metadata.dashboard	
cs.metadata.email_templates	
cs.metadata.export_jobs	
cs.metadata.export_templates	
cs.metadata.field_permissions	
cs.metadata.files	
cs.metadata.generic_template	
cs.metadata.images	
cs.metadata.import_jobs	
cs.metadata.model_metadatas	
cs.metadata.module_permissions	
cs.metadata.modules	
cs.metadata.pgfile	
cs.metadata.picklist_names	

```
at Object.callFunction (chrome-extension://bbcinkgjjkejfdpemiealijmmooekmp/onloadwff.js:71:745509)
at Object.handleRequest (chrome-extension://bbcinkgjjkejfdpemiealijmmooekmp/onloadwff.js:71:744677)
at d (chrome-extension://bbcinkgjjkejfdpemiealijmmooekmp/onloadwff.js:71:776740)
at e (chrome-extension://bbcinkgjjkejfdpemiealijmmooekmp/onloadwff.js:71:777605)
at e (chrome-extension://bbcinkgjjkejfdpemiealijmmooekmp/onloadwff.js:71:5465)
angular.js:15697
```



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